

Jared Marx-Kuo

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EMPLOYMENT	NSF Postdoc and Dickson Instructor , University of Chicago Math Department	2026 – Present
	Postdoctoral Visitor , Center for Mathematical Modeling, Santiago, Chile	June 2026
	NSF Postdoc , Rice University Math Department	2025 – 2026
	Griffith Conrad Evans Instructor , Rice University Math Department	2024 – 2025
EDUCATION	PhD in Mathematics , Stanford University	2019 – 2024
AWARDS	NSF Mathematical Sciences Postdoctoral Research Fellowship , National Science Foundation	2025 – 2028
	Rice Academy of Fellows , Rice University	2025 – 2026
	• Received \$2,000 professional development grant	
	Rice Creative Ventures Fund Grant , Rice University	October 2025
	• Received \$10,000 to host a conference on geometry and dynamics in April 2026	
	• Joint with David Fisher, Francisco Arana-Herrera, Christos Mantoulidis	
	US Junior Oberwolfach Fellow 2025 , MFO	July 2025
	• Received \$1000 to attend Oberwolfach workshop Differentialgeometrie im Großen 2025	
	Stanford CTL LIT Fellow , Stanford Center for Teaching and Learning	2023
	• \$2000 grant to develop TA handbook + workshops addressing: active learning, instructor talk, and identity in the classroom	
	• Joint with Alexandra Stavrianidi	
	NSF Graduate Research Fellow , National Science Foundation	2021
	• Full Funding for 3 years of PhD	
	Stanford Graduate Fellowship in Science & Engineering , Stanford University	2019
	• Full Funding for 3 years of PhD	
PUBLICATIONS AND PREPRINTS	Equivariant Allen–Cahn Solutions and the Existence of Cohomogeneity 2 Minimal Surfaces (Joint with Rayssa Caju and Pedro Gaspar) (2026, Submitted)	
	The p-widths of the Hemisphere (Accepted at Communications in Analysis and Geometry)	
	Mountain Pass Critical Points of the Volume Constrained Area Functional (Joint with Gregory R. Chambers) (In Revision)	
	Almgren’s Three-Legged Starfish (Joint with Christos Mantoulidis) (2025, To appear at the Proceedings of the AMS)	
	“Infinitely Many Surfaces with Prescribed Mean Curvature in the Presence of a Strictly Stable Minimal Surface” (Joint with Pedro Gaspar) (2025, Submitted)	
	“The p-widths of $\mathbb{RP}^2$” Proceedings of the AMS , Vol 153, No. 12 (2025) 5385 – 5396	
	“Diameter, Area, and Mean Curvature” (Joint with Gregory Chambers) Differential Geometry and its Applications , Vol 103 (2026)	
	“Index, Intersections, and Multiplicity of Min-Max Geodesics” (Joint with Lorenzo Sarnataro, Douglas Stryker) Advances in Mathematics , Volume 502, Part C. Article 111193	

- "The Isospectral Problem for p-widths: An Application of Zoll metrics"
Comptes Rendus Mathématique, Vol 363 (2025), 565 – 570
- "An Inverse Problem for Renormalized Area/Entanglement Entropy"
New Zealand Journal of Mathematics, Vol 56 (2025) 69 – 124
 (Video on this paper from BIRS 2024)
- "Second Inner Variations, Stress-Energy Tensors, and Index of Limiting Varifolds"
Journal of Geometric Analysis Vol 35, No. 290 (2025)
- "Geometric Variations of an Allen–Cahn Energy on Hypersurfaces" (Joint with Èrico Melo Silva)
Calculus of Variations and Partial Differential Equations, Vol 64, No. 180 (2025)
- "A Dirichlet-to-Neumann Map for the Allen-Cahn Equation on Manifolds with Boundary"
Interfaces and Free Boundaries, Vol 27, No. 4 (2025) 575 – 618
- "Variations of Renormalized Volume for Minimal Submanifolds of Poincare-Einstein Manifolds"
Communications in Analysis and Geometry, Vol 33, No. 1 (2025) 17 – 29
- "Sandpile Groups of Cayley Graphs of $\mathbb{F}_2^N$ " (Joint with Jiyang Gao, Vaughan McDonald, Chi Ho Yuen)
Communications in Algebra, Vol 52, No. 10 (2024), 4459 – 4479
- "Characters of Renner Monoids and Their Hecke Algebras" (Joint with Andy Hardt, Vaughan McDonald, John M. O'Brien, Alexander Vetter)
International Journal of Algebra and Computation, Vol 30, No. 7 (2020), 1505 – 1535
- "Convolution Algebras for Finite Reductive Monoids" (Joint with Vaughan McDonald, John M. O'Brien, Alexander Vetter)

TEACHING

- Math 355, Instructor**, *Linear Algebra*, Rice Jan 2025 – April 2025
- Math 354, Instructor**, *Honors Linear Algebra*, Rice Sept 2024 – Dec 2024
- Math 53 TA**, *Ordinary Differential Equations with Linear Algebra*, Stanford March 2023 – June 2023
- LSP Math Instructor**, *Leland Scholars program*, Stanford August 2021
- Instructor for course on high school level mathematics for incoming FGLI freshmen
 - Gave welcome talk, "*Mathematics and Society*"
- Math 62CM TA**, *Modern Mathematics: Continuous Methods*, Stanford January 2021 – March 2021
- Math 171 CA**, *Fundamental Concepts of Analysis*, Stanford September 2020 – November 2020
- SUMAC TA**, *Stanford University Mathematics Camp*, Stanford June 2020 – August 2020
- Teaching Assistant for summer course in algebraic topology
 - Gave talk, "*Math, Science, and Society*"

TALKS

- Recent Advances on Geometric Variational Problems and Curvature**, JMM Special Session January 2027
- Geometric Inequalities, Convergence, and Scalar Curvature**, JMM Special Session January 2027
- UMD Geometry Seminar**, University of Maryland College park October 2026
- PUC Geometry Seminar**, Pontificia Universidad Catolica de Chile June 2026
- IMPA Geometry Seminar**, Instituto Nacional de Matematica Pura e Aplicada June 2026
- CMM Analysis and PDE Seminar**, Center for Mathematical Modeling (Santiago, Chile) June 2026
- UTD Colloquium**, University of Texas Dallas March 2026
- Toronto Geometry Seminar**, University of Toronto March 2026
- Cal Tech Geometry Seminar**, California Institute of Technology February 2026
- UC Irvine Geometry Seminar**, University of California Irvine February 2026
- Yale Geometry Seminar**, Yale University February 2026
- BIP Inverse Problems Online Seminar**, (Hosted by Sebastian Munoz-Thon) October 2025
- UPenn Geometry Seminar**, University of Pennsylvania October 2025
- Lehigh University Geometry Seminar**, Lehigh University October 2025
- Rice University Undergraduate Colloquium**, Rice University October 2025
- Johns Hopkins University Geometry Seminar**, Johns Hopkins University September 2025
- Rutgers University Geometry Seminar**, Rutgers University September 2025
- Cornell University Geometry Seminar**, Cornell University September 2025

Oberwolfach Differentialgeometrie im Großen, MFO	<i>July 2025</i>
UC Santa Barbara Geometry Seminar , University of Santa Barbara California	<i>May 2025</i>
UC Santa Cruz Geometry and Analysis Seminar , University of Santa Cruz California	<i>May 2025</i>
Weyl Laws Across Mathematics , Brin Mathematics Research Center	<i>May 2025</i>
UTK Geometry Seminar , University of Tennessee Knoxville	<i>March 2025</i>
Conformal and CR Geometry Workshop , BIRS (Video)	<i>December 2024</i>
Rice Geometry Seminar , Rice University	<i>September 2024</i>
Cornell Geometry Seminar , Cornell University	<i>April 2024</i>
Syracuse Geometry Seminar , Syracuse University	<i>April 2024</i>
UChicago Geometry Seminar , University of Chicago	<i>April 2024</i>
Northwestern University Analysis Seminar , Northwestern University	<i>March 2024</i>
Calculus of Variations and Geometric Measure Theory Seminar , University of Pisa (Zoom)	<i>March 2024</i>
UC Davis Geometry Seminar , University of California Davis	<i>December 2023</i>
Rice Geometry Seminar , Rice University	<i>September 2023</i>
Early Career Seminar on Inverse Problems, PDEs, Microlocal Analysis, Differential Geometry , University of Washington	<i>March 2023</i>
UC Santa Cruz Geometry Seminar , University of California Santa Cruz	<i>November 2022</i>
UC Davis Geometry Seminar , University of California Davis	<i>October 2022</i>
UMD Geometry Seminar , University of Maryland College Park	<i>September 2022</i>
Stanford Geometry Seminar , Stanford University	<i>September 2022</i>
Conformal Geometry, Analysis, and Physics Conference in Honor of Robin Graham , University of Washington (Young researcher poster session)	<i>June 2022</i>

OUTREACH AND SERVICE	Referee service	(Periodic)
	• <i>Inventiones Mathematicae</i> • <i>Journal of Differential Geometry</i> • <i>Transactions of the American Mathematical Society</i> • <i>Selecta Mathematica</i> • <i>Nonlinear Analysis</i>	
	Thesis Committee Service , Yasser Chamaoui, Universite Libre de Bruxelles	September 2026
	Rice Summer Undergraduate Research Fellowship (SURF) Mentor Program Website • Individual reading with a student in Complex Analysis for 2 months	June-July 2025
	Oberwolfach Differentialgeometrie im Großen 2025 Reporter MFO • Workshop reporter for "Differentialgeometrie im Großen" 2025	July 2025
	Rice University Geometry Seminar Organizer ,	Sept 2024 - May 2026
	Rice University Math Ed Discussion Seminar , • Presented on <u>Identity and Privilege in the math classroom</u> • Led an interactive workshop discussing scenarios + actionable items using Universal Design for Learning (UDL) • Led a discussion on AI and how it is affecting homework and teaching in math classes	Dec 2024, Sept 2025
	Rice University Directed Reading Program Mentor, Program Website	September 2024 - June 2025
	Stanford TA Workshop, Program Website • Organized workshops on Active Learning, <u>Instructor Talk</u>, and <u>Identity in the Classroom</u>, to help TAs improve their teaching and student interactions • Sponsored by the Stanford CTL LIT Fellowship (see Awards)	Sept 2023 - June 2024
	Stanford Directed Reading Program Mentor, Program Website	September 2019 - June 2024
	Stanford Directed Reading Program Organizer, Program Website • Organized program and paired undergraduates with math graduate students for quarter-long, 1-on-1 reading projects • Advertise the program, schedule presentations, submit budget updates, etc.	January 2020 - June 2024
	President of Stanford Graduate Math Outreach Organization (SGMOO) , • Organizer for outreach programs run by math graduate students, including: Directed Reading Program (DRP), Stanford Women in Math Mentoring Program (SWIMM)	May 2022 - June 2024

- Stanford First-Year Mentoring Program Mentor + Organizer,** September 2020 - June 2023
- Paired older graduate students in math department with first year PhD students to facilitate bonding, research connections, and to provide PhD advice
 - Participated as mentee in 2019, as mentor in 2020-2023
- TA Mentor for First-time TAs** 2022-2023
- Mentored new TAs - observed their section teaching and provided feedback
 - Compensated role
- President of Stanford Math Graduate Students** September 2020 - September 2022
- Organized graduate student roles to develop a sense of community within the department
- SUMAC Application Reader, [Program Website](#)** March 2021 - April 2021
- Read applications for Stanford University Mathematics Camp
 - Advocated for underrepresented and underresourced students in the application process
- SURIM Mentor, [Stanford Undergraduate Research Institute in Mathematics](#)** June 2020 – August 2020
- Mentored undergraduates in PDEs with applications to the Allen-Cahn Equation
- SURA Mentor, [Program Website](#)** January 2020 - April 2020
- Mentored undergraduates in math by advising them on courses and summer projects
- FLI Community Mentor, [Program Website](#)** January 2020 - November 2020
- Supported low income undergraduate student with navigating college life and adjusting plans for Covid-19 pandemic

REFERENCES

Andre Neves, Professor, Department of Mathematics, University of Chicago,
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Otis Chodosh, Associate Professor, Department of Mathematics, Stanford University,
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Christos Mantoulidis, Associate Professor, Department of Mathematics, Rice University,
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